

Multiple Choice

1. **Answer:** D

Options: A) Eu; B) Gd; C) Am; D) Cu

Note: Copper (Cu) has an electron configuration of $[\text{Ar}] 3d^9 4s^1$, indicating its 4 f and 5 f orbitals are fully filled, unlike the lanthanides and actinides that possess partially filled 4 f or 5 f orbitals.

2. **Answer:** A

Options: A) 10 ml; B) 30 ml; C) 55 ml; D) 40 ml; E) 25 ml

Note: The correct answer is 10 ml because the stoichiometry of the neutralization reaction indicates that 2 moles of H_3PO_4 are required to neutralize 3 moles of $\text{Ca}(\text{OH})_2$, leading to the calculation that 0.025 M H_3PO_4 needs to be 10 ml to completely react with the 25 ml of 0.030 M $\text{Ca}(\text{OH})_2$.

3. **Answer:** A

Options: A) $1.6 \times 10^{-3}F$; B) $1.5 \times 10^{-3}F$; C) $1.4 \times 10^{-3}F$; D) $0.8 \times 10^{-3}F$; E) $0.9 \times 10^{-3}F$

Note: The correct answer is $1.6 \times 10^{-3}F$ because the formality is calculated by determining the moles of face

3. **Answer:** E

Options: A) .1002 : .3002 : .4002; B) .3582 : .4582 : .0582; C) .2582 : .4998 : .3092; D) .1582 : .6008 : .2092; E) .1582 : .4998 : .1092

Note: The correct answer reflects Graham's law of effusion, which states that the rate of diffusion of a gas is inversely proportional to the square root of its molar mass, resulting in the observed ratios for Ar, He, and Kr based on their respective molar masses.

4. **Answer:** B

Options: A) He_2^+ is the most stable among He , He_2 , and He_2^+ ; B) He_2^+ is more stable than He_2

Note: He_2^+ is more stable than He_2 because the removal of a π electron from He_2 reduces electron-electron repulsion and results in a greater bond order, while He is more stable than both due to its complete

True / False

6. **Answer:** True

Options: A) True; B) False

Note: The ionization energy of singly ionized helium is accurately calculated as 48.359 eV using the Bohr model due to its dependence on the effective nuclear charge and the formula for energy levels in hydrogen-like atoms, which reflects the increased attraction of the nucleus to the electron in helium.

7. **Answer:** True

Options: A) True; B) False

Note: At 12.19 K , an ideal gas can exist at 1 atm pressure and a concentration of 1 mole liter^{-1} because the ideal gas law ($PV = nRT$) holds true under these conditions, as long as the gas is not at a

7. **Answer:** False

Options: A) True; B) False

Note: $\text{Zn}(\text{NO}_3)_2$ is actually more soluble in water than NiSO_4 because nitrate ions (NO_3^-) are highly soluble in water, whereas the sulfate ion (SO_4^{2-}) tends to form less soluble compounds with certain cations.

8. **Answer:** False

Options: A) True; B) False

Note: The missing mass in ^{19}F , which corresponds to the mass defect due to binding energy, is not sign

8. **Answer:** False

Options: A) True; B) False

Note: The statement is false because the molality is calculated using the number of moles of solute (H_2SO_4) divided by the mass of the solvent (water) in kilograms, and with 49 g of H_2SO_4 corresponding to approximately 0.49 moles, the molality is actually 1.96 moles/kg , not 1.0 moles/kg .

Long Form Response

11. **Answer:** To dilute 5 gallons of a 90% ammonia solution to a 60% concentration, we need to calculate the volume of water required. First, we determine the amount of ammonia in the initial solution: $5\text{ gallons} \times 0.90 = 4.5\text{ gallons}$ of ammonia. Let x be the volume of water added. After dilution, the total volume will be $5 + x$ gallons, and we want the ammonia concentration to be 60% . The equation can be set up as follows: $4.5\text{ gallons of ammonia} / (5 + x)\text{ gallons} = 0.60$. Solving for x , we find that $4.5 = 0.60(5 + x)$, leading to $4.5 = 3 + 0.60x$, which simplifies to $1.5 = 0.60x$. Thus, $x = 2.5$ gallons. Therefore, 2.5 gallons of water are needed to achieve the desired 60% ammonia concentration.

Note: correct because it accurately applies the concept of dilution by calculating the initial amount of ammonia, setting up a proportion to find the final concentration, and solving for the volume of water needed to achieve the desired 60% concentration.

12. **Answer:** The expansion of one cubic foot of air when heated from 0°C to $46,000^\circ\text{C}$ can be analyzed through the equation $PV = nRT$. As temperature increases significantly, the volume of the gas expands exponentially, assuming pressure is constant. *Note:* correct because it applies the Ideal Gas Law, illustrating that the significant increase in temperature

End of Answer Key